



AuditAI: Transforming Financial Auditing Through Intelligent Automation

Senior Project CPCS 499

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Project Overview

The Platform



AuditAI is an intelligent, rule-based automated financial audit tool designed to validate financial records across five distinct audit dimensions.

Core Capabilities

- ✓ Automates time-consuming financial reconciliation.
- ✓ Replaces manual data cross-referencing with OCR-based validation.
- ✓ Classifies detected discrepancies into meaningful technical categories.
- ✓ Provides structured outputs via an interactive web dashboard and Excel reports.

Problem Definition: The Constraints of Traditional Auditing



Time-Consuming Manual Effort

Financial reconciliation relies on repetitive, manual cross-checking between highly fragmented data sources (sales, receipts, AP, AR).



Unstructured Data Blindspots

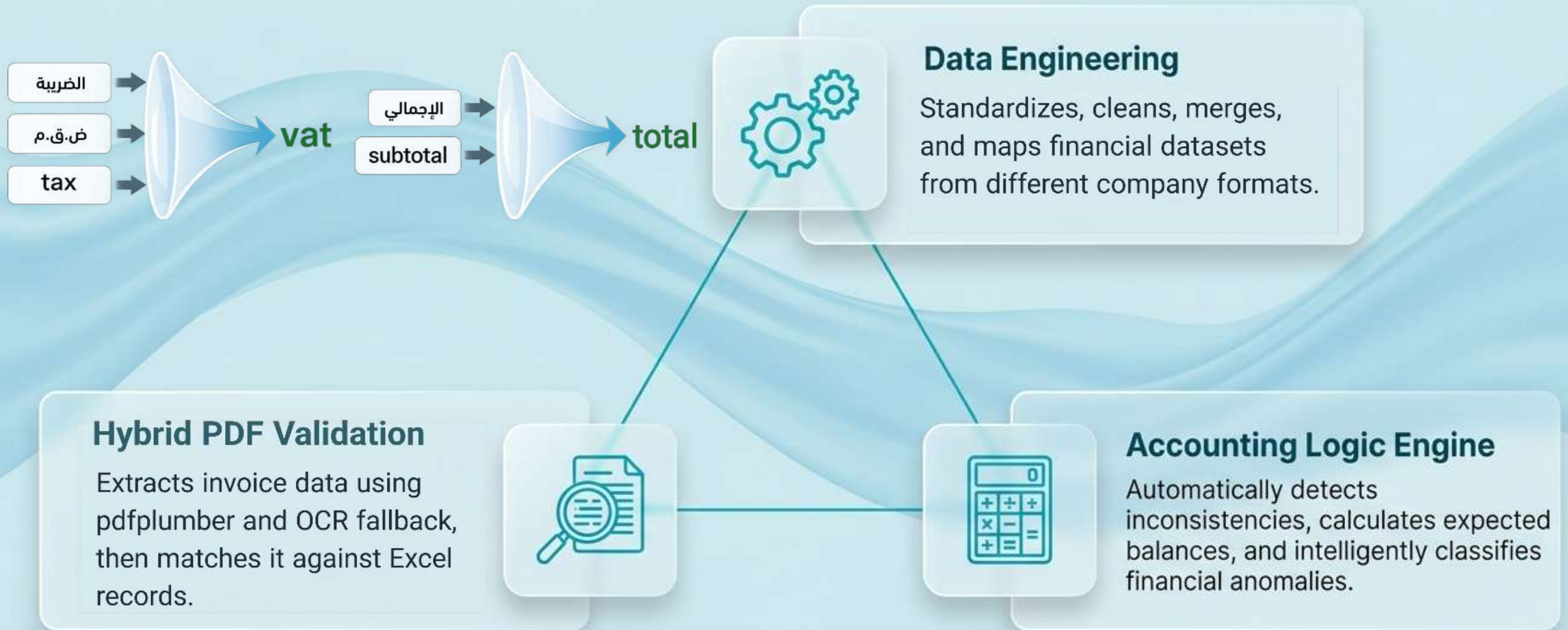
External auditors struggle to efficiently extract and analyze large volumes of unstructured data, particularly localized PDF invoices.



High Risk of Human Error

Manual data entry and complex cross-referencing inevitably lead to undetected anomalies and misclassified financial exceptions.

Proposed Solution: Structuring & Validating Financial Data



Modular 3-Tier System Architecture

Presentation Layer

- Java Script web application.
- Handles auditor UI, authentication, client management, and interactive results visualization.



Presentation Layer

Business Logic Layer

- Python-based core audit engine.
- Contains 4 independent audit modules, the Smart Column Detector, and the OCR Validation pipeline.



Business Logic Layer

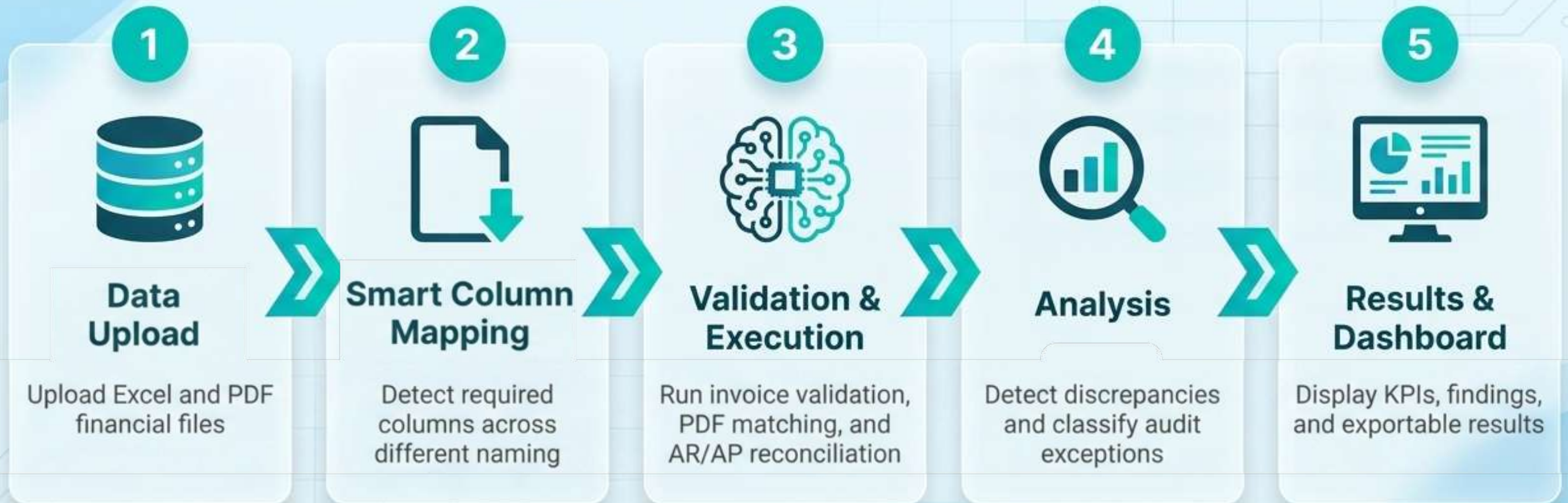
Data Layer

- Supabase for user authentication and relational data management.
- File-based input system supporting 10 financial document types (Excel/PDF).
- Outputs structured JSON (dashboard) and Excel files.



Data Layer

End-to-End System Workflow



Five-Dimension Audit Logic

Part	Name	Core Question	Method
1	Sales Invoice Audit	Are all sales invoices correct?	Duplicate / Gap / VAT check
2	Purchase Invoice Audit	Are all purchase orders correct?	Duplicate / Gap / VAT check
3	AR Reconciliation	Does each customer's balance make sense?	Opening + Sales - Receipts = Closing
4	AP Reconciliation	Does each supplier's balance make sense?	Opening + Purchases - Payments = Closing
5	PDF Validation	Does the PDF match the Excel record?	Hybrid pdflumber + OCR engine

Example for Sales/Purchases Invoice Audit

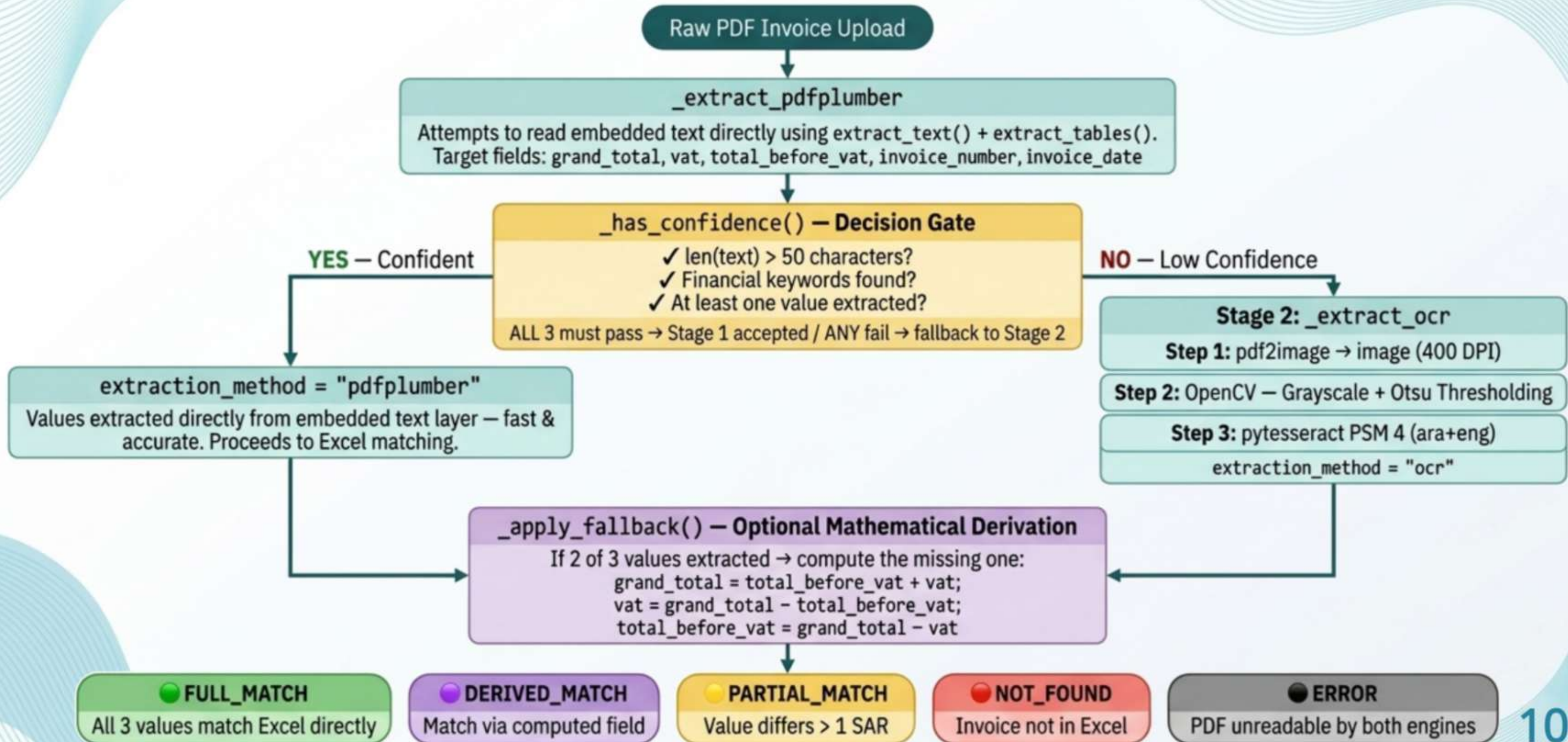
Invoice #	Customer	Total Before VAT	VAT Recorded	Result
1001	Al Falafil Rest.	1,000.00 SAR	150.00 SAR	OK
1001	Al Falafil Rest.	500.00 SAR	75.00 SAR	DUPLICATE
1003	Al Nakheel Co.	800.00 SAR	80.00 SAR	VAT ERROR
1004	Frozen Mart	2,000.00 SAR	300.00 SAR	OK
—	(gap)	—	—	MISSING

Example for AR/AP Reconciliation

Customer	Opening AR	+ Sales	– Receipts	Expected Closing	Actual Closing	Result
Al Falafil Rest.	10,000	50,000	45,000	15,000	15,000	OK
Al Nakheel Co.	8,000	30,000	28,000	10,000	13,000	Mismatch
Frozen Mart	0	20,000	20,000	0	0	OK

Supplier	Opening AP	+ Purchases	– Payments	Expected Closing	Actual Closing	Result
Al Marai Co.	20,000	80,000	75,000	25,000	25,000	OK
Nadec Foods	8,000	40,000	35,000	13,000	16,500	Mismatch
Albaik Supply	0	15,000	15,000	0	0	OK

Hybrid PDF Extraction Engine



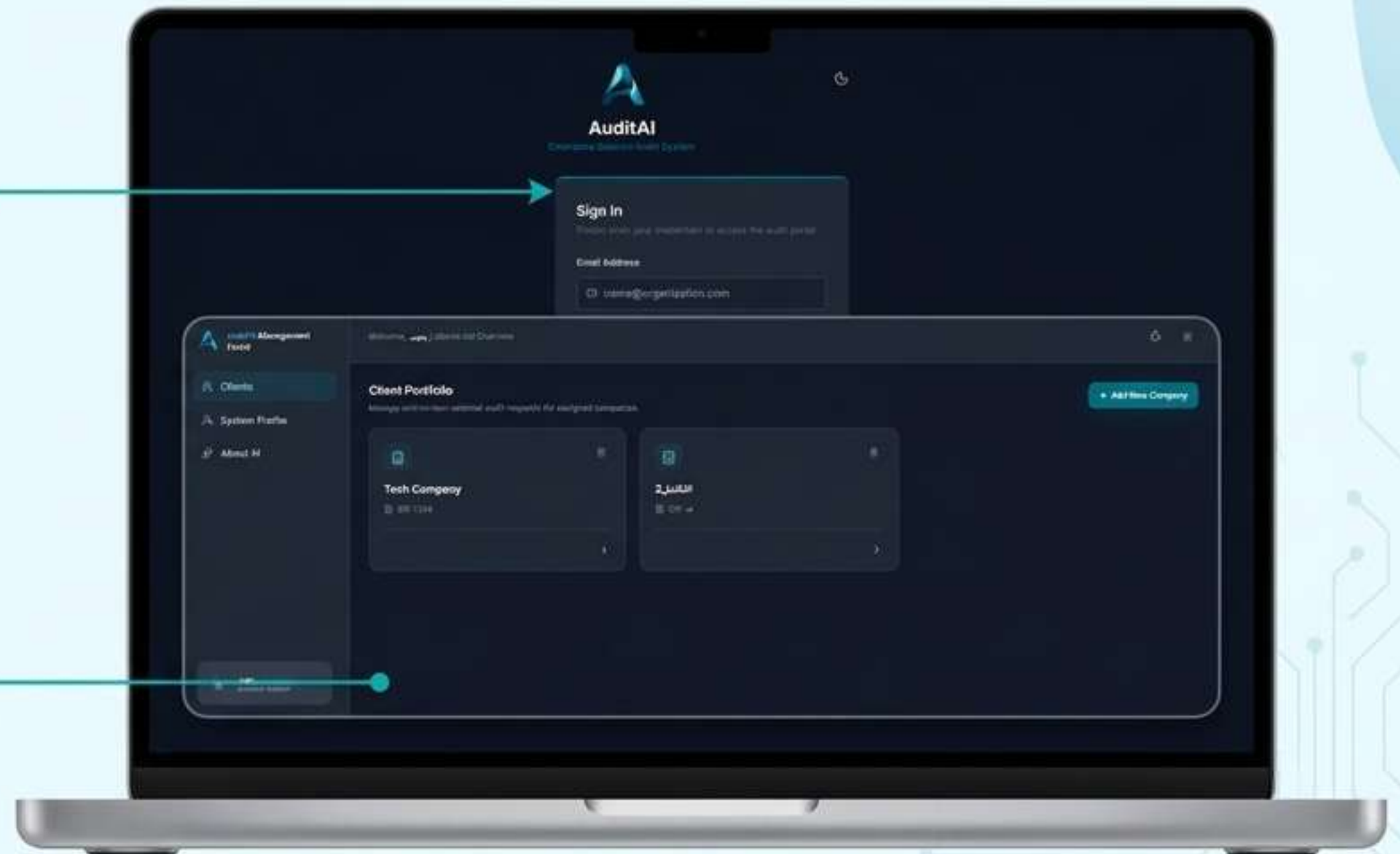
System Tour: Secure Access & Portfolio Management

Secure Access

Role-based authentication (email/password and Google OAuth) with encrypted connections.

Client Portfolio

Centralized dashboard allowing the external auditor to manage multiple company profiles, assign CR numbers, and track specific financial audit periods (fiscal years).





AuditAI

Enterprise External Audit System

Sign In

Please enter your credentials to access the audit portal

Email Address

name@organization.com

Password

•••••



Sign In →

OR

 Sign in with Google

Don't have an account? [Sign up](#)

[Forgot Password?](#)

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AuditAI Management Portal



Clients



System Profile



About AI



يحيى

External Auditor

Welcome, يحيى | [clients-list Overview](#)



Client Portfolio

Manage and monitor external audit requests for assigned companies.

[+ Add New Company](#)



Tech Company

CR: 1234



التكامل 2

CR: ٢٠٢٤



System Tour: Unified Upload Module

Centralized Data Ingestion

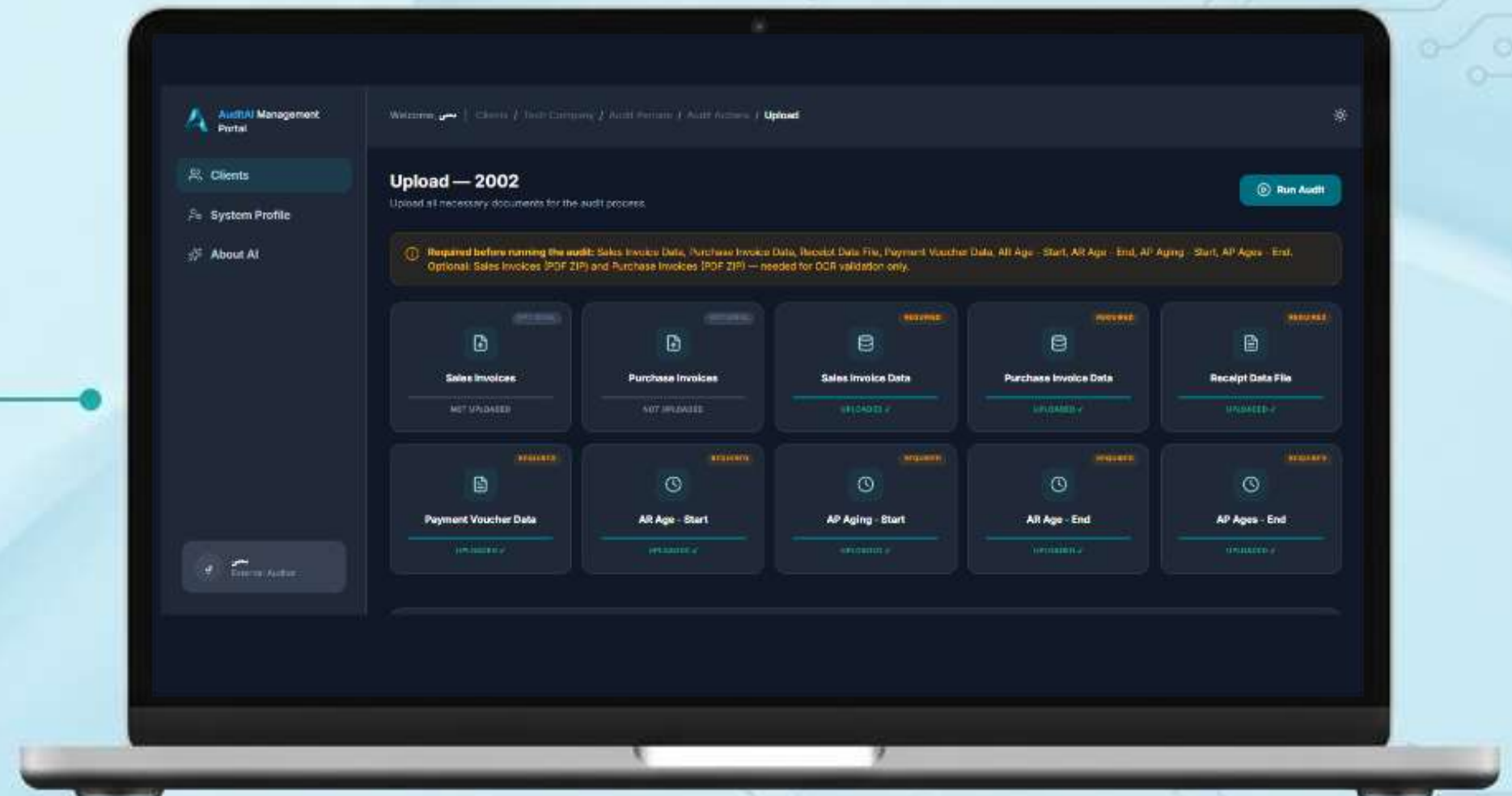
- A drag-and-drop interface capable of accepting up to 10 distinct financial datasets simultaneously.

Smart Column Detection

- Backend logic automatically maps user-uploaded files to required system variables, regardless of the client's original formatting.

Dynamic Validation

- Visually flags 'Required' vs 'Optional' documents (e.g., PDF ZIP files required for OCR validation only).



System Tour: Dashboard & Analytics

Real-Time Execution

Python analytical logic executes automatically upon data upload, passing results via Supabase to the frontend.

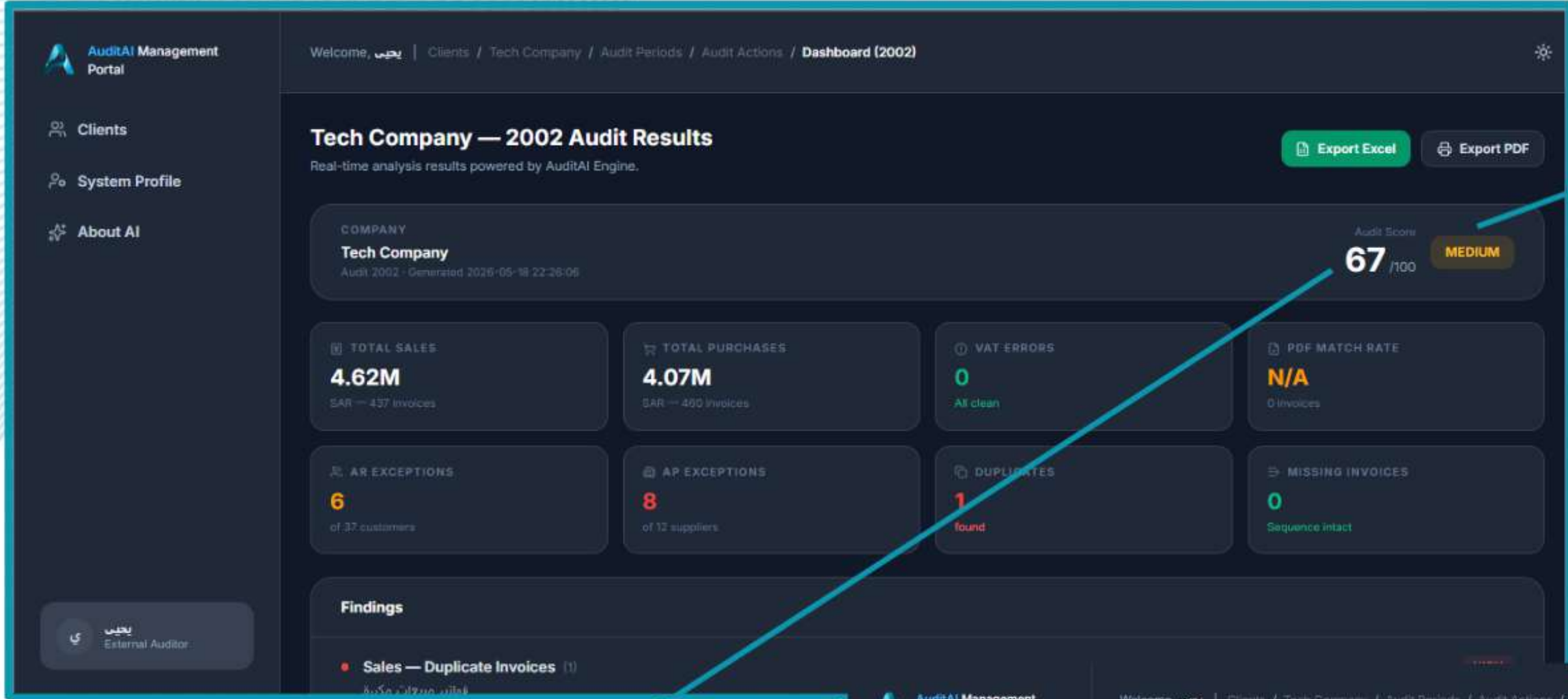
Technical Findings

Categorizes discrepancies by Type (e.g., Missing Invoices, VAT Errors, AR Mismatch), Count, and Severity (High Priority, Warning).

Interactive Visualization

KPI tracking, overall audit scoring (e.g., 80/100 Medium Risk), and direct export to Excel for final reporting.





Risk Level

Score	Level
85–100	LOW
65–84	MEDIUM
40–64	HIGH
0–39	CRITICAL

Audit Score

Issue	Deduction	Cap
VAT error	-2 per error	-20 max
Duplicate invoice	-3 per duplicate	-15 max
Missing invoice	-1 per gap	-10 max
AR exception	-2 per customer	-10 max
AP exception	-2 per supplier	-10 max
OCR mismatch	-1 per 10% miss rate	-10 max



- Clients
- System Profile
- About AI

AR Exceptions

CUSTOMER	OPENING AR	SALES	RECEIPTS	EXPECTED	ACTUAL	DIFFERENCE
مطعم عمر العربي	67,548	593,209	507,944	152,812	119,300	+33,512
شركة يونيك المحدودة	0	1,441,592	1,074,879	366,712	339,006	+27,707
شركة اديم المستقبل التجاريه	0	213,029	175,206	37,823	35,014	+2,809
بوفيه بدره علي يحيى البيشي	1,440	0	0	1,440	0	+1,440
كافيتريا القريات	0	2,186	852	1,334	0	+1,334
مؤسسه شفق الطهي	0	18,078	16,848	1,231	805	+426

AP Exceptions

SUPPLIER	OPENING AP	PURCHASES	PAYMENTS	EXPECTED	ACTUAL	DIFFERENCE
شركة الخليج الغربيه للاستيراد المحدوده	0	0	249,506	-249,506	52,024	-301,530

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External Auditor

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شركة اكسال العالميه للتجاره	0	0	73,888	-73,888	16,813	-90,701
شركة وزد العربيه	0	617,173	618,230	196,943	184,105	+14,838
شركة بنده للتجزئه	0	47,931	42,494	5,437	0	+5,437
مركز شعاعه للتمويل	24,620	41,752	65,728	644	0	+644
شركة ساجر العربيه التجاريه	104	0	0	104	0	+104
شركة رواد للتسويق المحدوده	0	6,101	6,100	1	0	+1

Technical Findings

PART	ISSUE TYPE	DESCRIPTION	COUNT	SEVERITY
Sales	Duplicate Invoices	فواتير مبيعات مكرره	1	HIGH
AR	Reconciliation Gap	فروقات في مطابقة الاعم المدينه	6	MEDIUM
AP	Reconciliation Gap	فروقات في مطابقة الاعم الدائنه	8	MEDIUM

يحيى
External Auditor

Rigorous multi-level testing ensures academic and real-world reliability

Unit Testing

Mathematically verified financial equations and individual Python data processing functions.

Integration Testing

Confirmed seamless data flow from OCR extraction, through the Supabase database, to the frontend dashboard.

System Testing

Evaluated overall stability and accuracy by processing massive, real-world financial datasets.

Edge Case Handling

Proven robustness against zero-value transactions, missing data fields, non-sequential invoices, and completely unbalanced ledgers without system failure.

Tools & Technologies



A modern, responsive web stack bridging Python-driven data engineering with a scalable JavaScript presentation layer.

Empirical Results & System Validation

Metrics Dashboard

438

Total real-world PDF invoices successfully processed during final testing.

100%

Extraction success rate across both the pdfplumber and pytesseract OCR fallback engines.

96.6%

Overall Match Rate.

- Zero false positives returned in VAT validation, invoice sequencing, and duplicate detection tests.
- Final automated Audit Score calculated accurately: 80/100 (Medium Risk level).

Conclusion & Future Work

Conclusion

- Successfully automated financial validation and reconciliation tasks
- Reduced manual auditing effort and improved audit accuracy
- Integrated invoice matching, PDF validation, AR/AP reconciliation, and web-based audit management into one platform.

Future Work

- NLP-based audit recommendations
- Integration with cloud accounting systems
- Improved automatic data cleaning and column detection



Thank You

We invite the evaluation panel for questions.